

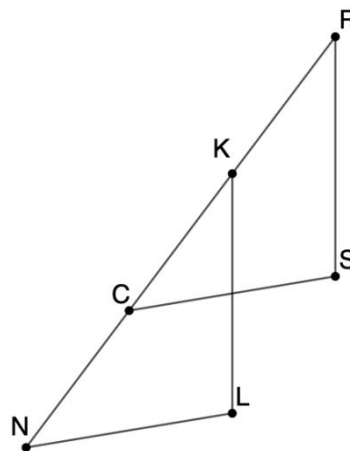
Geometry
Unit 6
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Given: $\overline{NL} \parallel \overline{CS}$ and $\angle KRS$ and $\angle LKR$ are supplementary

Prove: $\triangle NKL \sim \triangle CRS$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
	$\overline{NL} \parallel \overline{CS}$	Given
1.	$\angle KNL \cong \angle RCS$	If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
2.	$\angle KRS$ and $\angle LKR$ are supplementary angles	Given
3.	$\overline{KL} \parallel \overline{RS}$	If consecutive interior angles are supplementary, then the two lines intersected by a transversal are parallel.
4.	$\angle NKL \cong \angle CRS$	If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
5.	$\triangle NKL \sim \triangle CRS$	AA ~

$\triangle NKL \sim \triangle CRS$

AA ~

$\angle KRS$ and $\angle LKR$ are supplementary angles

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

$\angle NKL \cong \angle CRS$

If consecutive interior angles are supplementary, then two lines intersected by a transversal are parallel.

$\overline{KL} \parallel \overline{RS}$

Reflexive Property of Congruence

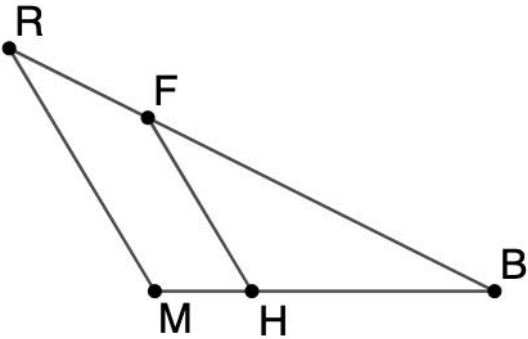
$\angle KNL \cong \angle RCS$

Given

Given: $RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

Prove: $\triangle RBM \sim \triangle FBH$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
6.	$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$	Given
7.	$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$	Division Property of Equality
8.	$\frac{RF}{RB} = \frac{MH}{MB}$	Substitution Property of Equality
9.	$\angle RBM \cong \angle FBH$	Reflexive Property of Congruence
10.	$\triangle RBM \sim \triangle FBH$	SAS ~

$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$

$\triangle RBM \sim \triangle FBH$

$\frac{RF}{RB} = \frac{MH}{MB}$

$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

$\angle RBM \cong \angle FBH$

SAS ~

Given

Reflexive Property of Congruence

Substitution Property of Equality

Division Property of Equality